

BIOS640_week3_finalproject

Sarfo Okyere Kore

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#Exercise 1 ## Keep adults and set clear category order ### Import the cleaned dataset

```
nhanes <- read_csv("cleaned_NHANES.csv", show_col_types = FALSE)

nhanes_adults <- nhanes |>
  filter(age >= 18) |>
  mutate(
    gender = factor(gender, levels = c("Male", "Female")),
    ethnicity_2 = factor(
      ethnicity_2,
      levels = c(
        "Mexican",
        "Other Hispanic",
        "White",
        "Black",
        "Asian",
        "Other"
      )
    )
  )
```

Figure_1: Age distribution by gender

```
figure_1 <- ggplot(
  nhanes_adults,
  aes(x = age, fill = gender)
) +
  geom_histogram(
    binwidth = 5,
    boundary = 17.5,
    position = "dodge",
    colour = "white"
  ) +
  labs(
    title = "Age Distribution by Gender",
    x = "Age (years)",
    y = "Number of participants",
    fill = "Gender"
  ) +
  scale_fill_manual(values = c("Male" = "#F8766D", "Female" = "#00BFC4")) +
```

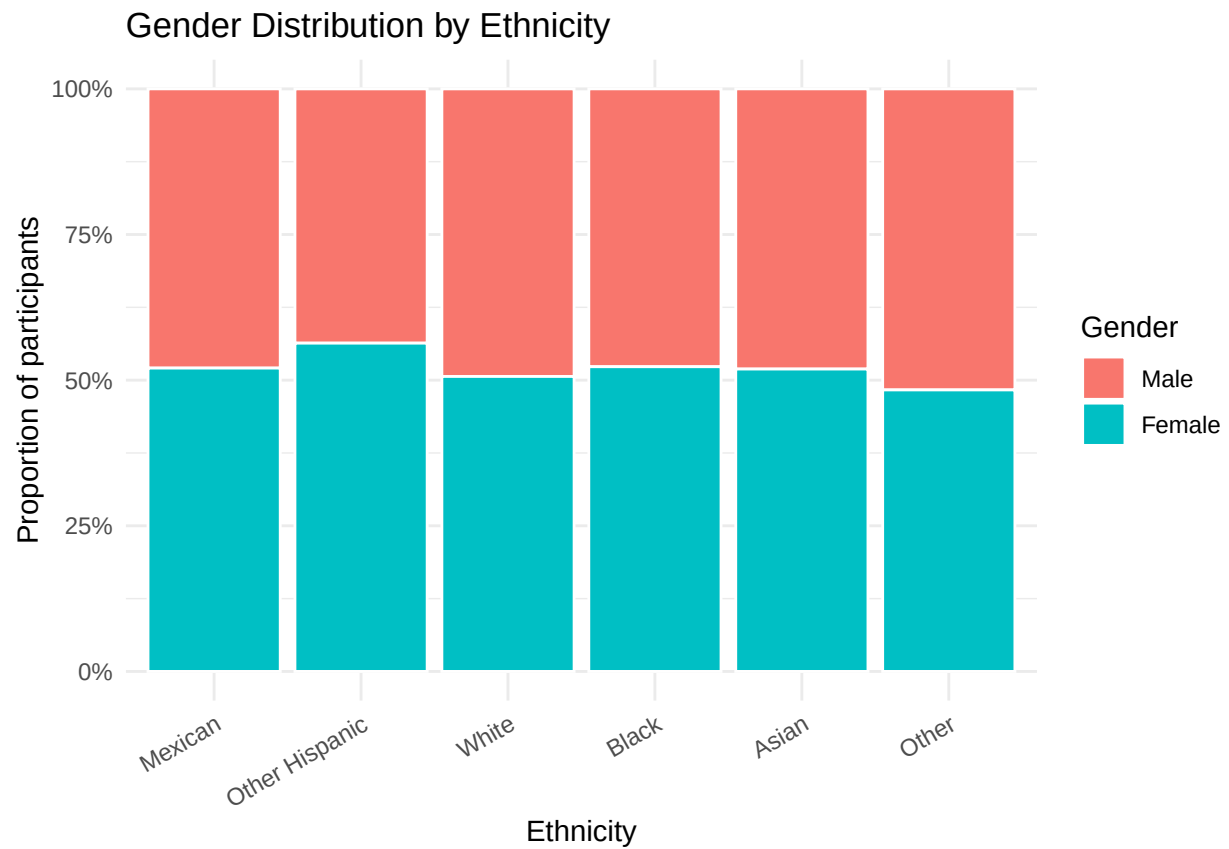
```
theme_minimal()
```

figure_1



```
##Figure_2: : Gender composition within ethnicity categories
figure_2 <- ggplot(
  nhanes_adults,
  aes(x = ethnicity_2, fill = gender)
) +
  geom_bar(position = "fill", colour = "white") +
  scale_y_continuous(labels = percent_format()) +
  scale_fill_manual(values = c("Male" = "#F8766D", "Female" = "#00BFC4")) +
  labs(
    title = "Gender Distribution by Ethnicity",
    x = "Ethnicity",
    y = "Proportion of participants",
    fill = "Gender"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 30, hjust = 1)
  )
```

figure_2

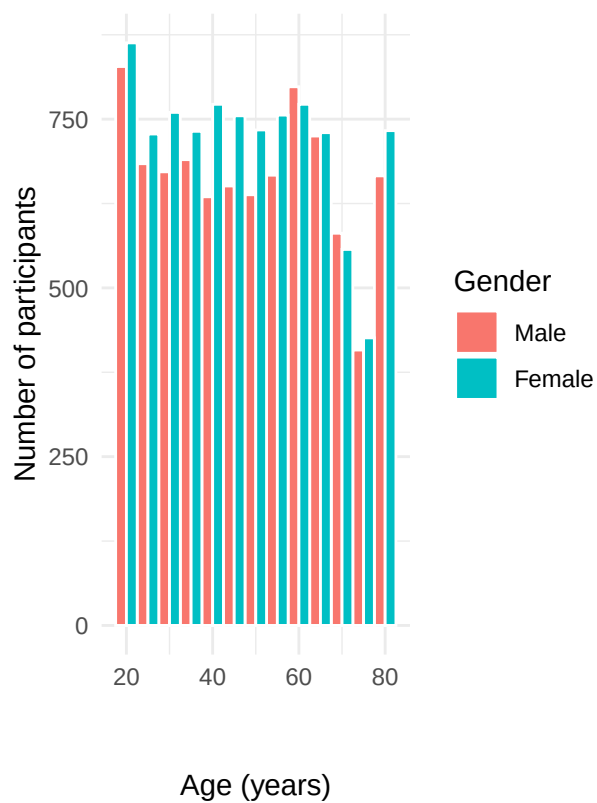


##Combining the two graphs using cowplot

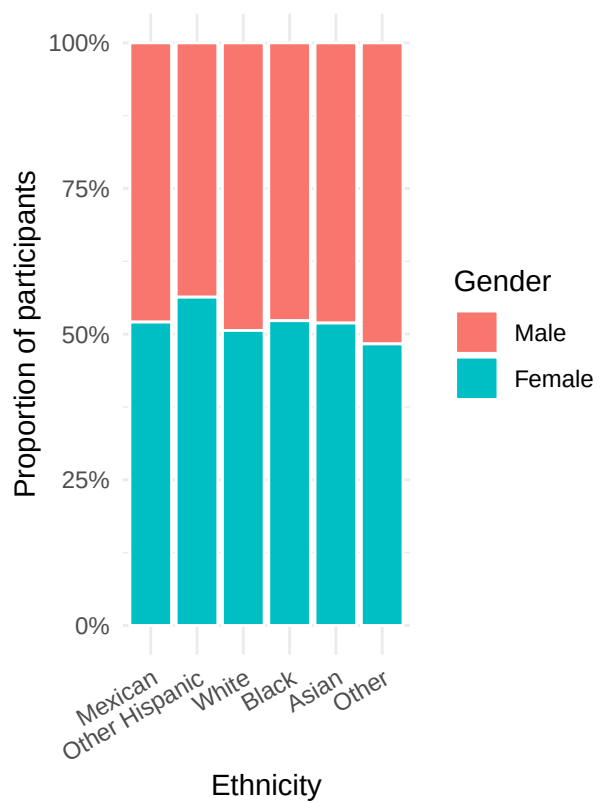
```
combined_cowplot <- plot_grid(  
  figure_1,  
  figure_2,  
  nrow = 1,  
  labels = c("A", "B"),  
  align = "h"  
)
```

```
combined_cowplot
```

A Age Distribution by Gender



B Gender Distribution by Ethnicity



##combining the graphs with ggarrange

```
library(ggpubr)
```

```
combined_ggpubr <- ggarrange(  
  figure_1,  
  figure_2,  
  ncol = 1,  
  nrow = 2,  
  labels = c("A", "B"),  
  align = "v"  
)
```

```
combined_ggpubr
```



`##plot_grid()` seems to refer to simply placing two plots next to each other for visual inspection. It's a straightforward way to see differences but may lack polish in alignment or spacing. `##ggarrange()` Provides a more structured layout. It lets you control the arrangement (rows, columns), add labels (A, B, etc.), and ensure consistent sizing.

`##Exercise 2`

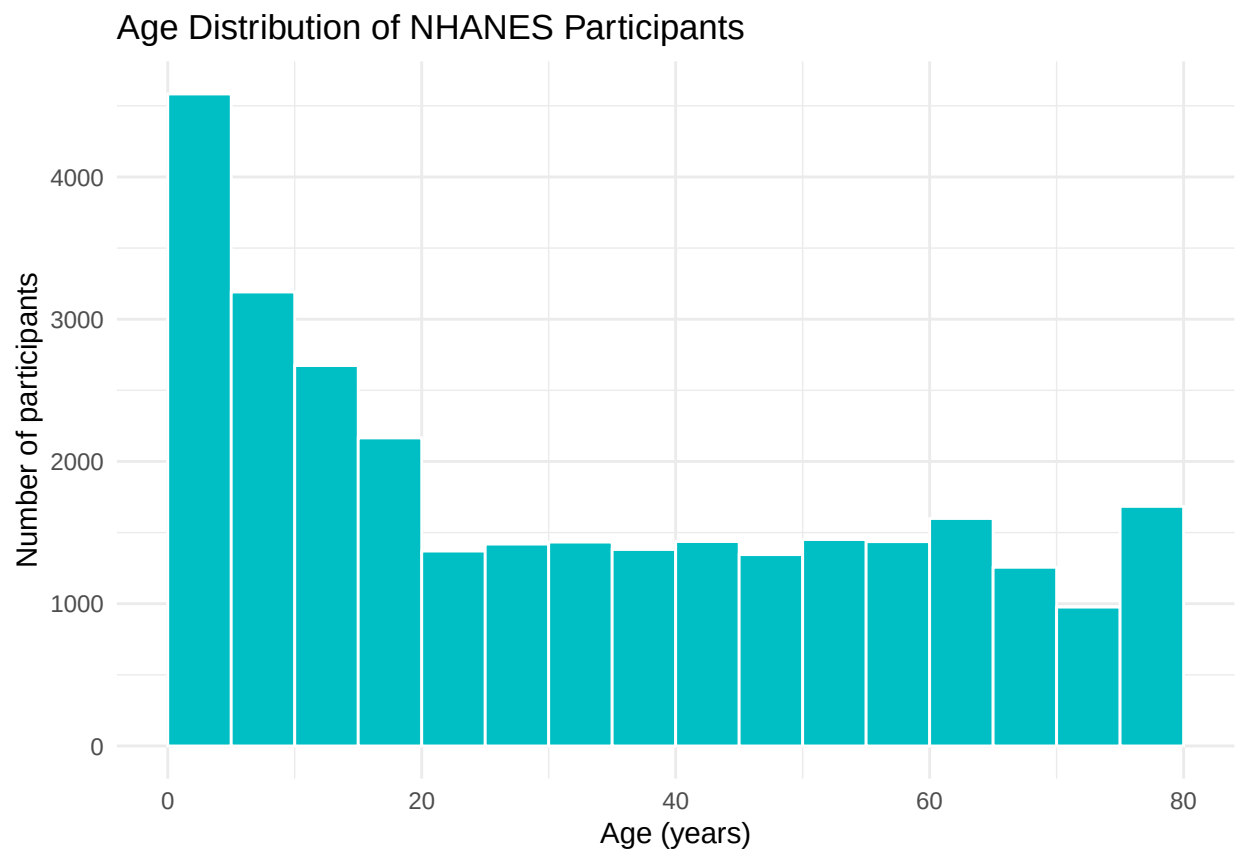
```
##Creating appropriate plots for age
###Summary statistics and missing values for age
age_summary <- nhanes |>
  summarise(
    n = sum(!is.na(age)),
    missing = sum(is.na(age)),
    mean_age = mean(age, na.rm = TRUE),
    median_age = median(age, na.rm = TRUE),
    sd_age = sd(age, na.rm = TRUE),
    min_age = min(age, na.rm = TRUE),
    max_age = max(age, na.rm = TRUE)
  )

age_summary
```

```
## # A tibble: 1 x 7
##       n missing mean_age median_age sd_age min_age max_age
##   <int>   <int>   <dbl>     <dbl> <dbl>   <dbl>   <dbl>
## 1 29400     0    32.5       28   24.9     0     80
```

```
## Age distribution plot
plot_age <- ggplot(nhanes, aes(x = age)) +
  geom_histogram(
    binwidth = 5,
    boundary = 0,
    fill = "#00BFC4",
    colour = "white"
  ) +
  labs(
    title = "Age Distribution of NHANES Participants",
    x = "Age (years)",
    y = "Number of participants"
  ) +
  theme_minimal()

plot_age
```



```
## Save the plot
ggsave(
  filename = "figures/age_distribution.png",
  plot = plot_age,
  width = 7,
  height = 5,
  dpi = 300
)
```

```
## Ensure logical ordering of the recoded categories
nhanes <- nhanes |>
  mutate(
    gender = factor(gender, levels = c("Male", "Female")),
    ethnicity_1 = factor(
      ethnicity_1,
      levels = c("Mexican", "Other Hispanic", "White", "Black", "Other")
    ),
    ethnicity_2 = factor(
      ethnicity_2,
      levels = c(
        "Mexican", "Other Hispanic", "White",
        "Black", "Asian", "Other"
      )
    )
  )
)
)
###Gender
gender_summary <- nhanes |>
  count(gender) |>
  mutate(percent = n / sum(n) * 100)

gender_missing <- sum(is.na(nhanes$gender))

gender_summary
```

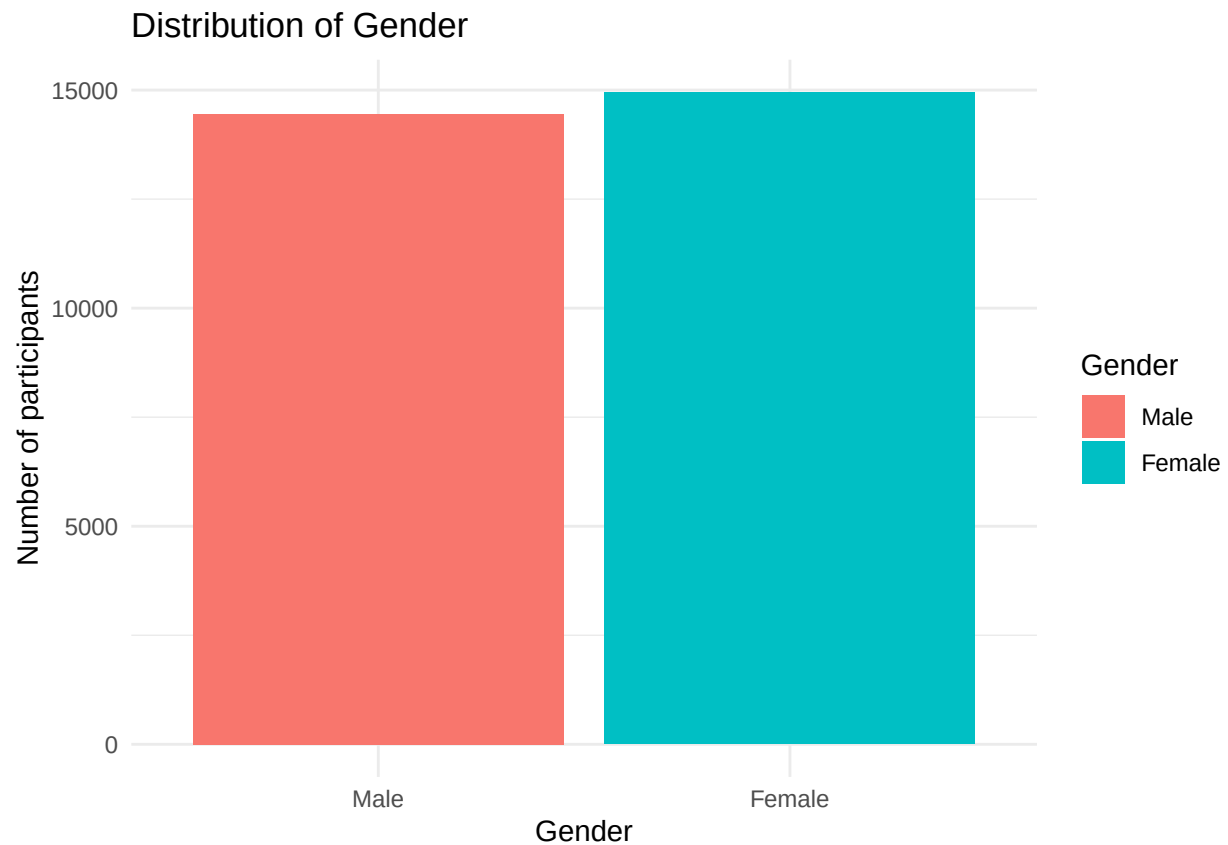
```
## # A tibble: 2 x 3
##   gender      n percent
##   <fct> <int>   <dbl>
## 1 Male   14452    49.2
## 2 Female 14948    50.8
```

```
gender_missing
```

```
## [1] 0
```

```
plot_gender <- ggplot(nhanes, aes(x = gender, fill = gender)) +
  geom_bar() +
  labs(
    title = "Distribution of Gender",
    x = "Gender",
    y = "Number of participants",
    fill = "Gender"
  ) +
  scale_fill_manual(values = c("Male" = "#F8766D", "Female" = "#00BFC4")) +
  theme_minimal()

plot_gender
```



```
ggsave(
  filename = "figures/gender_distribution.png",
  plot = plot_gender,
  width = 7,
  height = 5,
  dpi = 300
)
```

```
ggsave(
  filename = "figures/gender_distribution.png",
  plot = plot_gender,
  width = 7,
  height = 5,
  dpi = 300
)
```

```
##First ethnicity variable
ethnicity_1_summary <- nhanes |>
  count(ethnicity_1) |>
  mutate(percent = n / sum(n) * 100)

ethnicity_1_missing <- sum(is.na(nhanes$ethnicity_1))

ethnicity_1_summary
```

```
## # A tibble: 5 x 3
```



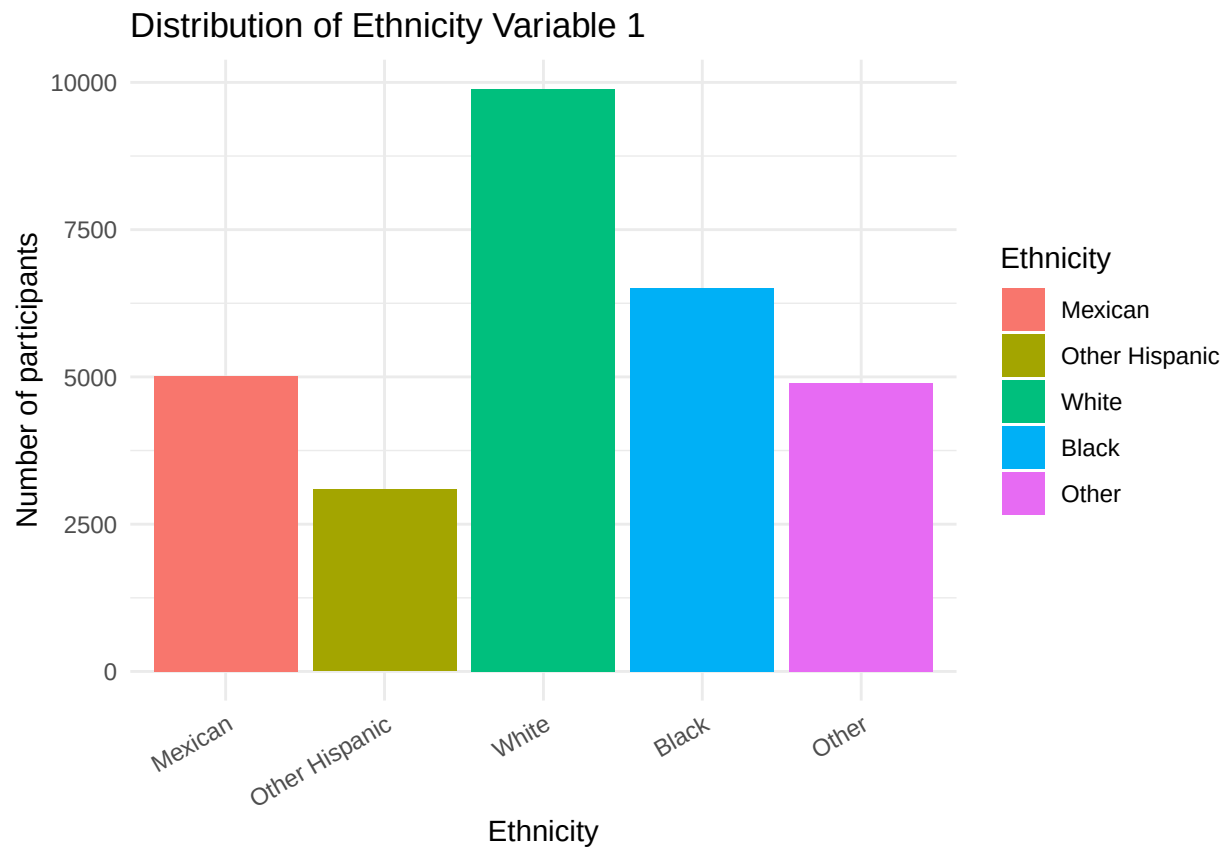
```
## ethnicity_1      n percent
## <fct>          <int>  <dbl>
## 1 Mexican      5018   17.1
## 2 Other Hispanic 3088   10.5
## 3 White         9890   33.6
## 4 Black         6511   22.1
## 5 Other         4893   16.6
```

```
ethnicity_1_missing
```

```
## [1] 0
```

```
plot_ethnicity_1 <- ggplot(nhanes, aes(x = ethnicity_1, fill = ethnicity_1)) +
  geom_bar(show.legend = TRUE) +
  labs(
    title = "Distribution of Ethnicity Variable 1",
    x = "Ethnicity",
    y = "Number of participants",
    fill = "Ethnicity"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 30, hjust = 1)
  )
```

```
plot_ethnicity_1
```



```

ggsave(
  filename = "figures/ethnicity_1_distribution.png",
  plot = plot_ethnicity_1,
  width = 8,
  height = 5,
  dpi = 300
)

```

```

### Ethnicity variable 2
ethnicity_2_summary <- nhanes |>
  count(ethnicity_2) |>
  mutate(percent = n / sum(n) * 100)

ethnicity_2_missing <- sum(is.na(nhanes$ethnicity_2))

ethnicity_2_summary

```

```

## # A tibble: 6 x 3
##   ethnicity_2      n percent
##   <fct>         <int>   <dbl>
## 1 Mexican         5018    17.1
## 2 Other Hispanic  3088    10.5
## 3 White          9890    33.6
## 4 Black           6511    22.1
## 5 Asian           3284    11.2
## 6 Other           1609     5.47

```

```

ethnicity_2_missing

```

```

## [1] 0

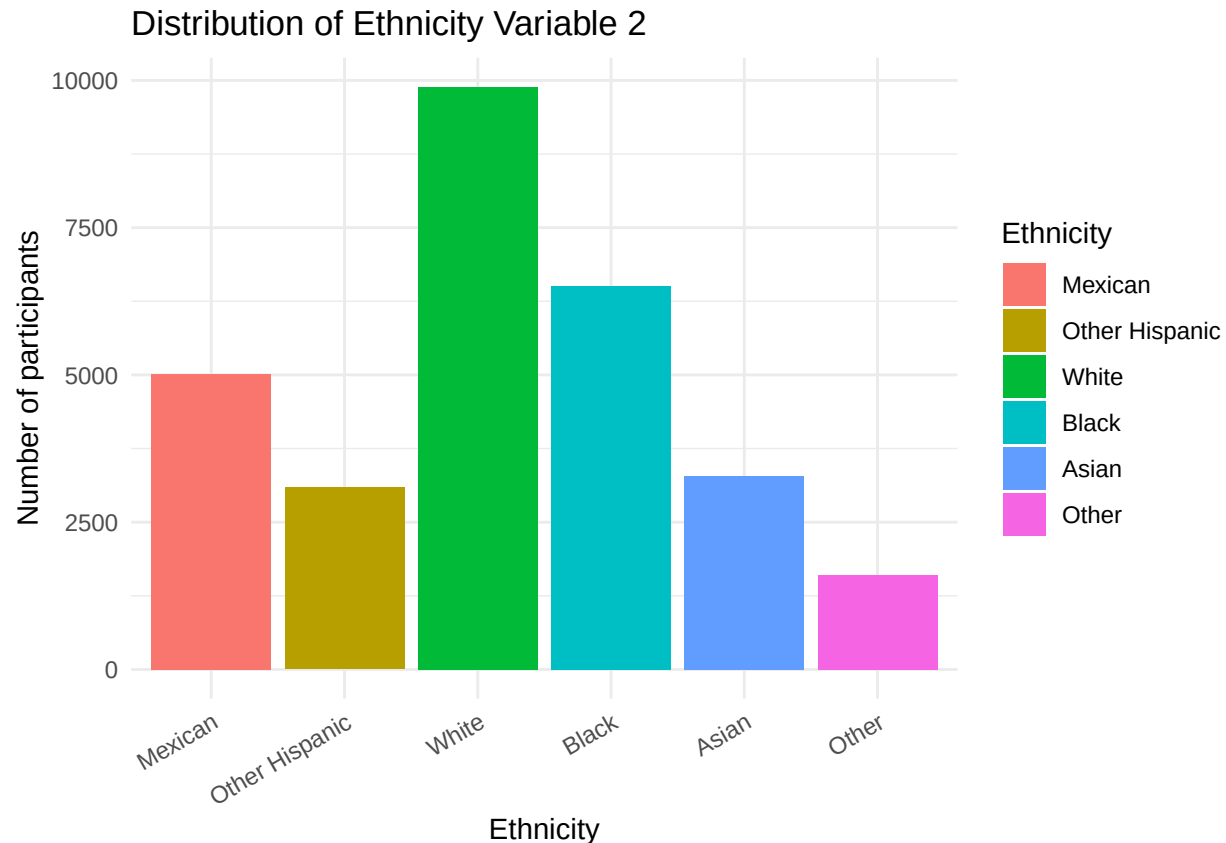
```

```

plot_ethnicity_2 <- ggplot(nhanes, aes(x = ethnicity_2, fill = ethnicity_2)) +
  geom_bar(show.legend = TRUE) +
  labs(
    title = "Distribution of Ethnicity Variable 2",
    x = "Ethnicity",
    y = "Number of participants",
    fill = "Ethnicity"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 30, hjust = 1)
  )

plot_ethnicity_2

```



```
ggsave(
  filename = "figures/ethnicity_2_distribution.png",
  plot = plot_ethnicity_2,
  width = 8,
  height = 5,
  dpi = 300
)

# taking out the less detailed ethnicity variable
nhanes <- nhanes |>
  select(-ethnicity_1)
```

There are no missing values in gender, ethnicity_1, or ethnicity_2. The gender counts are very similar (14,452 Male; 14,948 Female). White participants are the largest category in both ethnicity variables. ## I will keep ethnicity_2 because it retains the separate Asian category (3,284 participants), whereas ethnicity_1 combines this group into Other. It therefore preserves more information.

Exercise 3 ## My colleague's plot shows raw body weight, which makes comparisons across participants difficult because they start at very different baseline weights. It also uses many colours and a long numeric legend, creating visual clutter without making individual trajectories easier to follow. The title is also missing, the axis labels are vague, and there is no visual reference for each participant's baseline weight.

A clearer approach is to calculate each participant's change in weight from week 0. Every participant will therefore start at zero; negative values indicate weight loss and positive values indicate weight gain. Faceting gives every participant a separate panel, avoiding overlapping lines and eliminating the need for an unreadable colour legend. Keeping the same y-axis scale across panels also allows direct comparison of the magnitude of change

```

## Import the diet dataset
diet <- read_csv("diet.csv", show_col_types = FALSE)

## Extract each participant's baseline weight at week 0
baseline <- diet |>
  filter(Week == 0) |>
  select(Participant, baseline_weight = Weight)

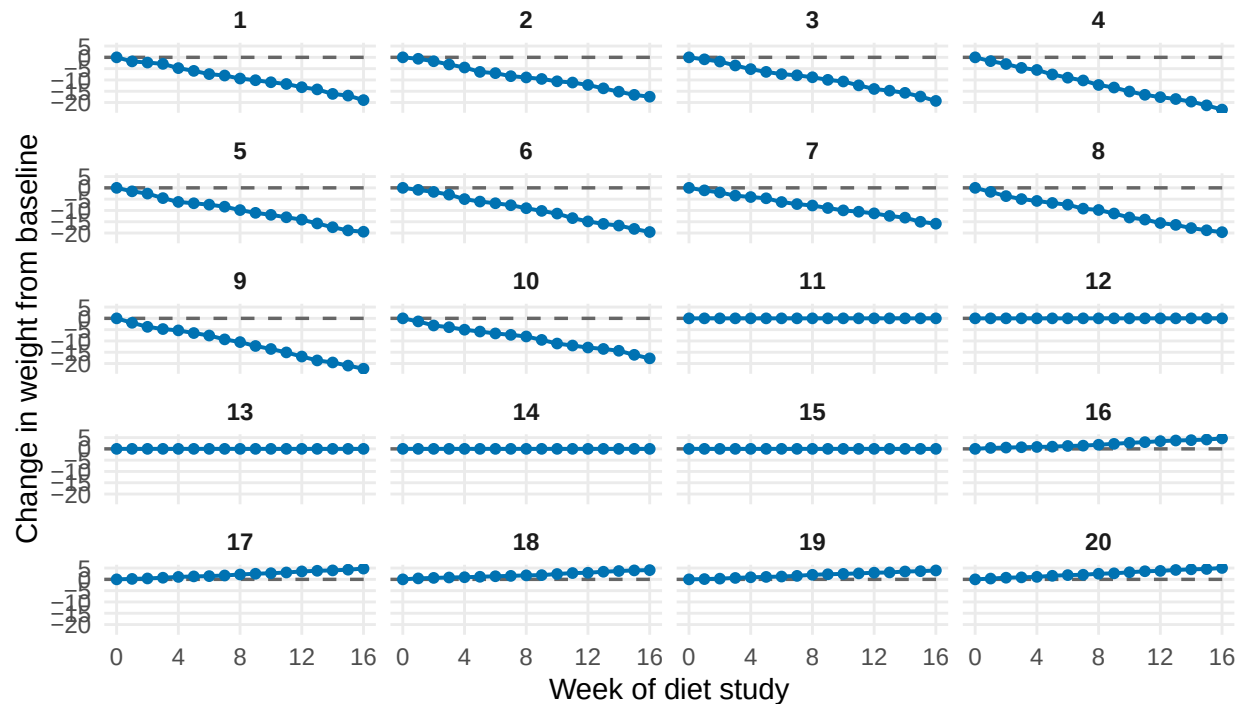
## Calculate change in weight from baseline
diet_change <- diet |>
  left_join(baseline, by = "Participant") |>
  mutate(
    change_from_baseline = Weight - baseline_weight,
    Participant = factor(Participant)
  )

### Improved figure
plot_diet <- ggplot(
  diet_change,
  aes(x = Week, y = change_from_baseline, group = Participant)
) +
  geom_hline(
    yintercept = 0,
    colour = "grey40",
    linetype = "dashed",
    linewidth = 0.6
  ) +
  geom_line(
    colour = "#0072B2",
    linewidth = 0.7
  ) +
  geom_point(
    colour = "#0072B2",
    size = 1.4
  ) +
  facet_wrap(~ Participant, ncol = 4) +
  scale_x_continuous(breaks = seq(0, 16, by = 4)) +
  labs(
    title = "Change in Participant Weight During the Diet Study",
    subtitle = "Each participant's weight change is measured relative to their week 0 baseline",
    x = "Week of diet study",
    y = "Change in weight from baseline",
    caption = "Values below zero indicate weight loss; the dashed line represents baseline."
  ) +
  theme_minimal() +
  theme(
    strip.text = element_text(face = "bold"),
    plot.title = element_text(face = "bold"),
    panel.grid.minor = element_blank()
  )
plot_diet

```

Change in Participant Weight During the Diet Study

Each participant's weight change is measured relative to their week 0 baseline



Values below zero indicate weight loss; the dashed line represents baseline.

#Exercise 4

```
## Average available SBP readings for each participant
nhanes$average_sbp <- rowMeans(
  nhanes[, c(
    "systolic_bp_1", "systolic_bp_2",
    "systolic_bp_3", "systolic_bp_4"
  )],
  na.rm = TRUE
)

## rowMeans returns NaN when all readings are missing
nhanes$average_sbp[is.nan(nhanes$average_sbp)] <- NA_real_
```

```
## Keep the final analytic sample
final_sample <- nhanes |>
  filter(
    !is.na(age),
    !is.na(gender),
    !is.na(average_sbp)
  ) |>
  mutate(
    gender = factor(gender, levels = c("Male", "Female")),
    age_cat = factor(
      age_cat,
```

```

    levels = c("0-17", "18-24", "25-39", "40-59", "60+")
  ),
  sbp_category = case_when(
    average_sbp < 120 ~ "Normal",
    average_sbp < 130 ~ "Elevated",
    average_sbp < 140 ~ "Stage 1 hypertension",
    average_sbp >= 140 ~ "Stage 2 hypertension"
  ),
  sbp_category = factor(
    sbp_category,
    levels = c(
      "Normal",
      "Elevated",
      "Stage 1 hypertension",
      "Stage 2 hypertension"
    )
  )
)
)
)

```

```

# Report the sample-size change
tibble(
  original_sample = nrow(nhanes),
  final_sample = nrow(final_sample),
  excluded = nrow(nhanes) - nrow(final_sample)
)

```

```
## # A tibble: 0 x 3
```

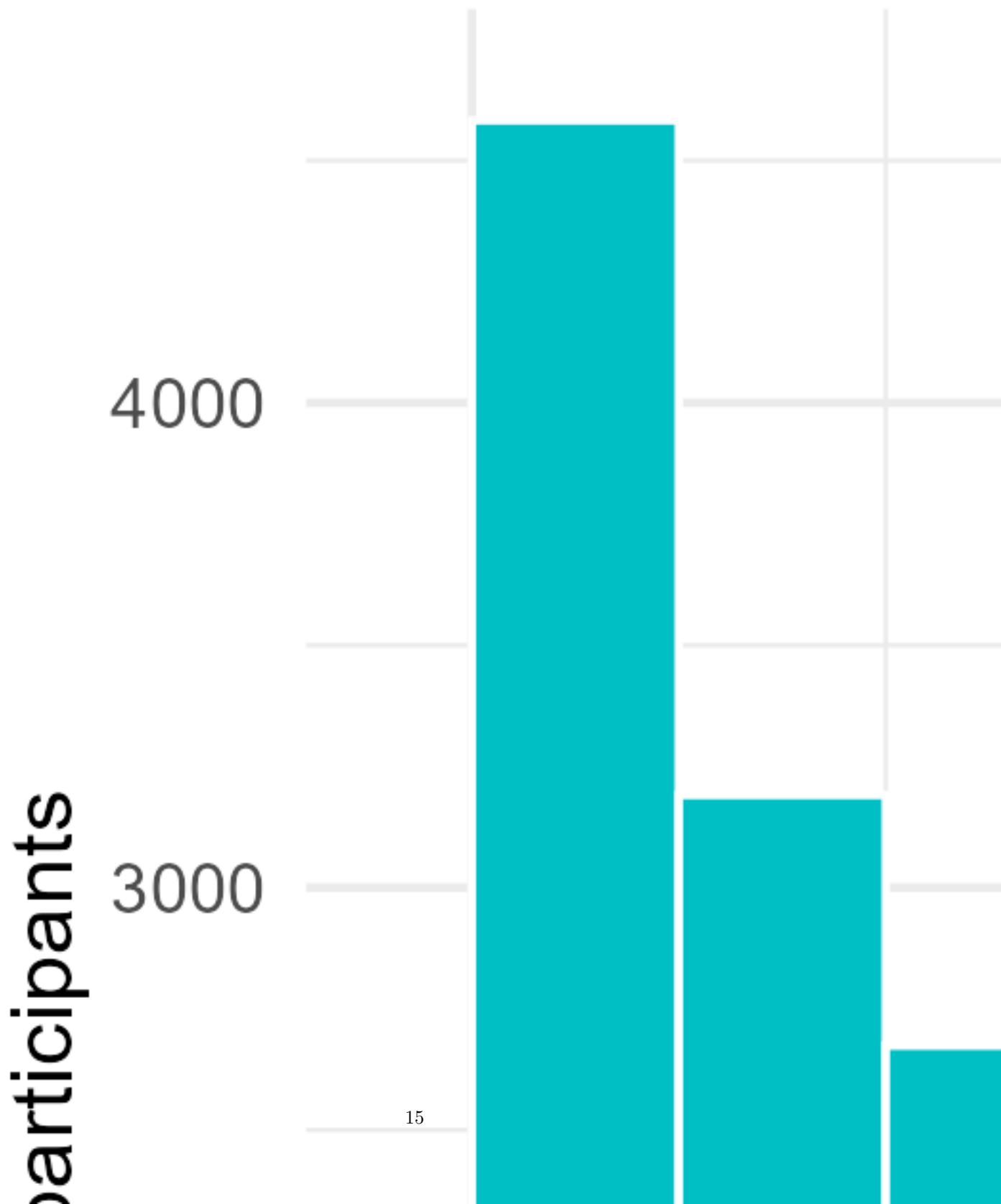
```
## # i 3 variables: original_sample <int>, final_sample <int>, excluded <int>
```

```

###importing my saved age distribution from exercise 2
knitr::include_graphics("figures/age_distribution.png")

```

Age Distribution

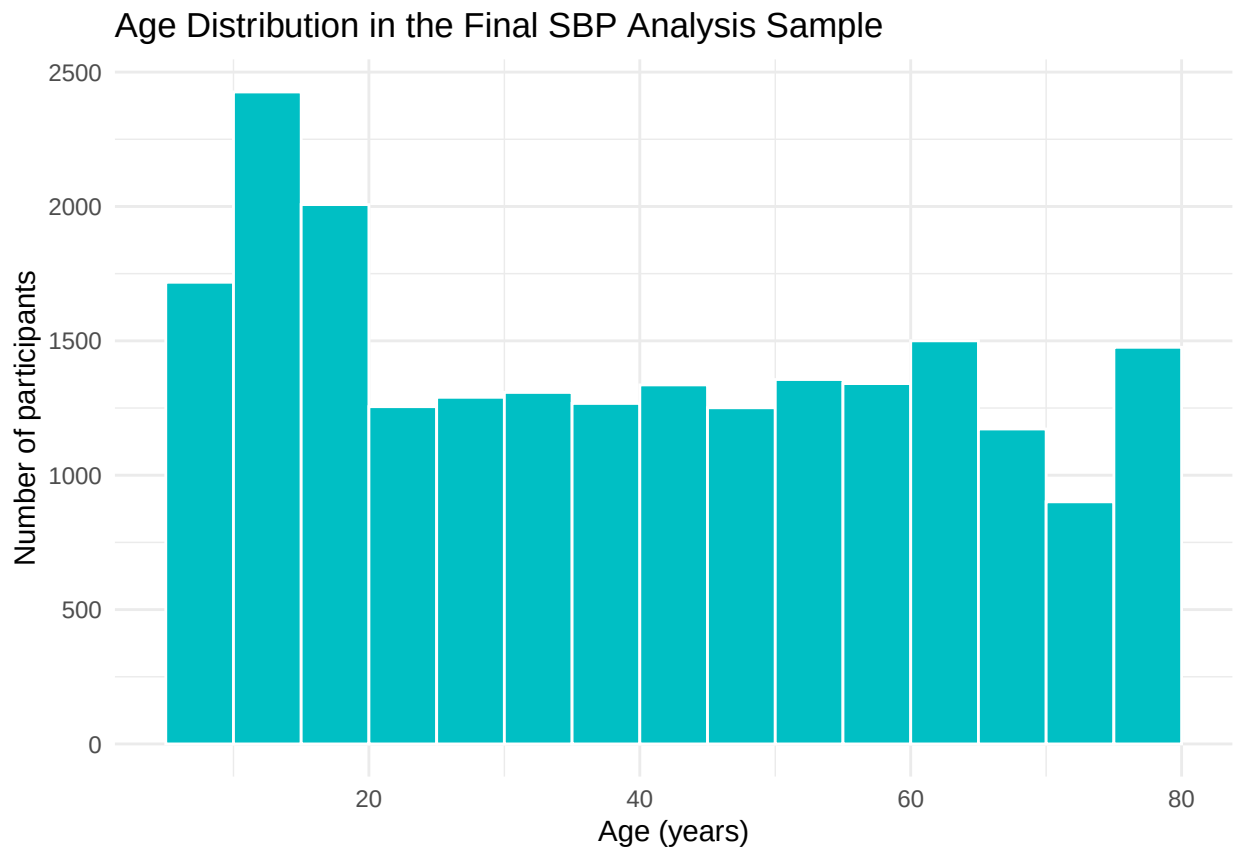


```

###create the equivalent plot for the final sample
plot_age_final <- ggplot(final_sample, aes(x = age)) +
  geom_histogram(
    binwidth = 5,
    boundary = 0,
    fill = "#00BFC4",
    colour = "white"
  ) +
  labs(
    title = "Age Distribution in the Final SBP Analysis Sample",
    x = "Age (years)",
    y = "Number of participants"
  ) +
  theme_minimal()

plot_age_final

```



```

###Hypertension-category counts
plot_sbp_counts <- ggplot(final_sample, aes(x = sbp_category, fill = sbp_category)) +
  geom_bar(show.legend = FALSE) +
  scale_fill_manual(
    values = c(
      "Normal" = "#009E73",
      "Elevated" = "#F0E442",
      "Stage 1 hypertension" = "#E69F00",

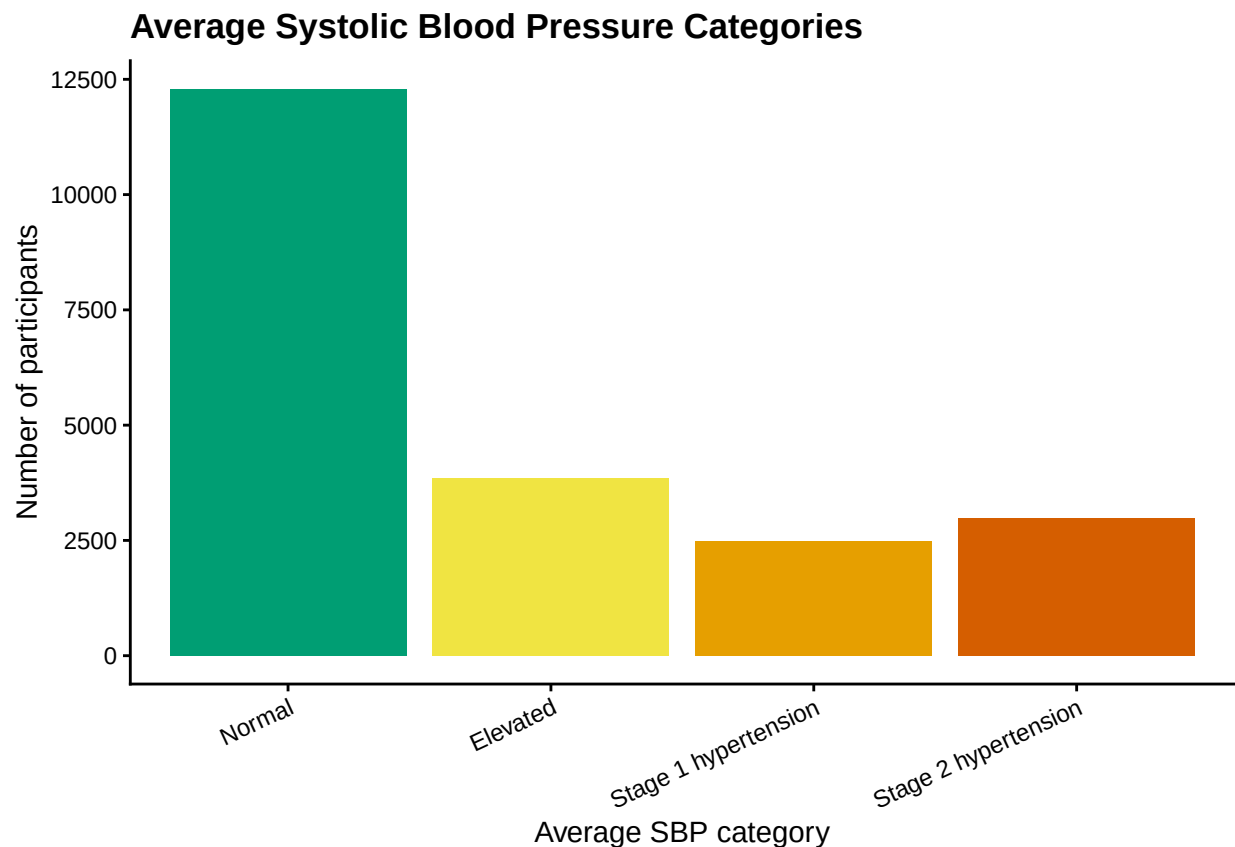
```



```

    "Stage 2 hypertension" = "#D55E00"
  )
) +
labs(
  title = "Average Systolic Blood Pressure Categories",
  x = "Average SBP category",
  y = "Number of participants"
) +
theme_classic() +
theme(
  plot.title = element_text(face = "bold"),
  axis.text.x = element_text(angle = 25, hjust = 1)
)
plot_sbp_counts

```



```

### Gender distribution in the final sample
plot_gender_final <- ggplot(final_sample, aes(x = gender, fill = gender)) +
  geom_bar(show.legend = FALSE) +
  scale_fill_manual(
    values = c("Male" = "#0072B2", "Female" = "#CC79A7")
  ) +
labs(
  title = "Gender Distribution in the Final SBP Analysis Sample",
  x = "Gender",

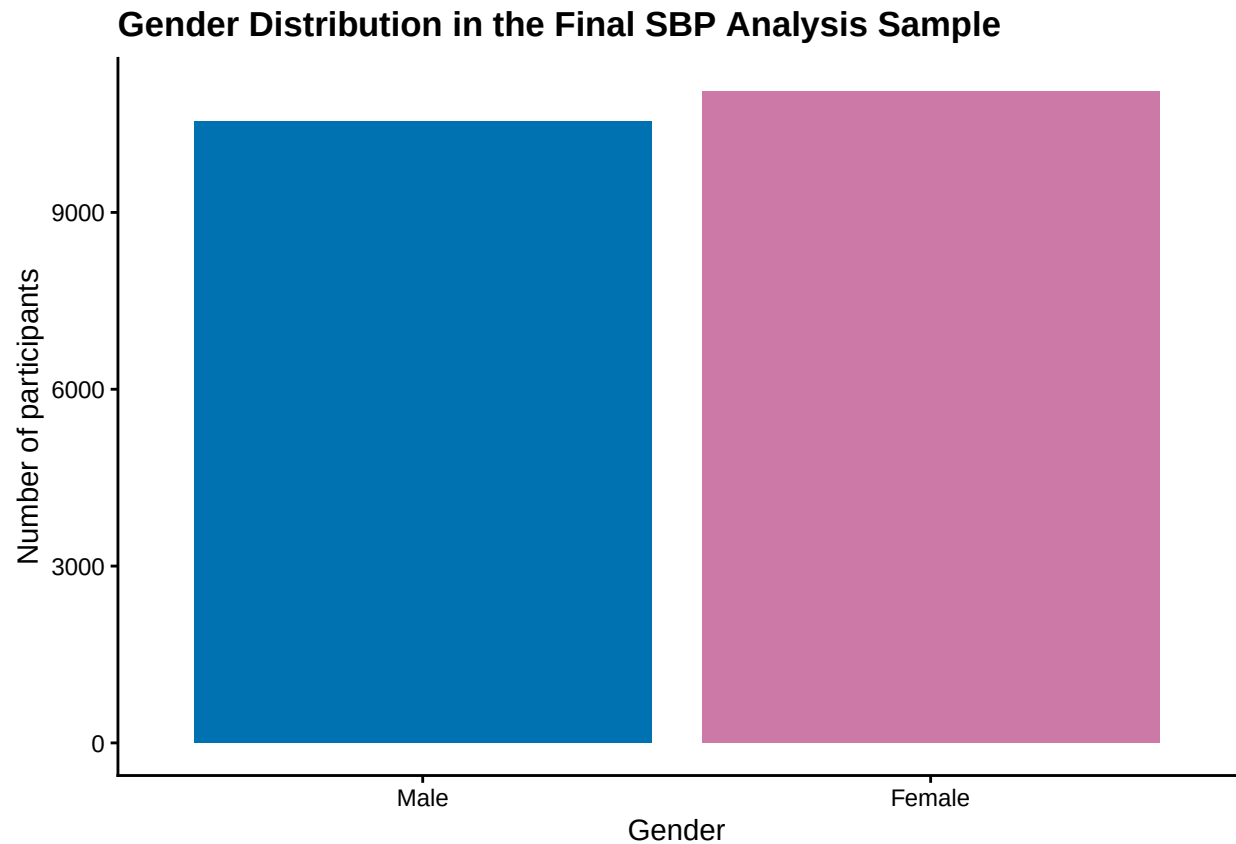
```

```

  y = "Number of participants"
) +
theme_classic() +
theme(
  plot.title = element_text(face = "bold")
)

```

plot_gender_final



```

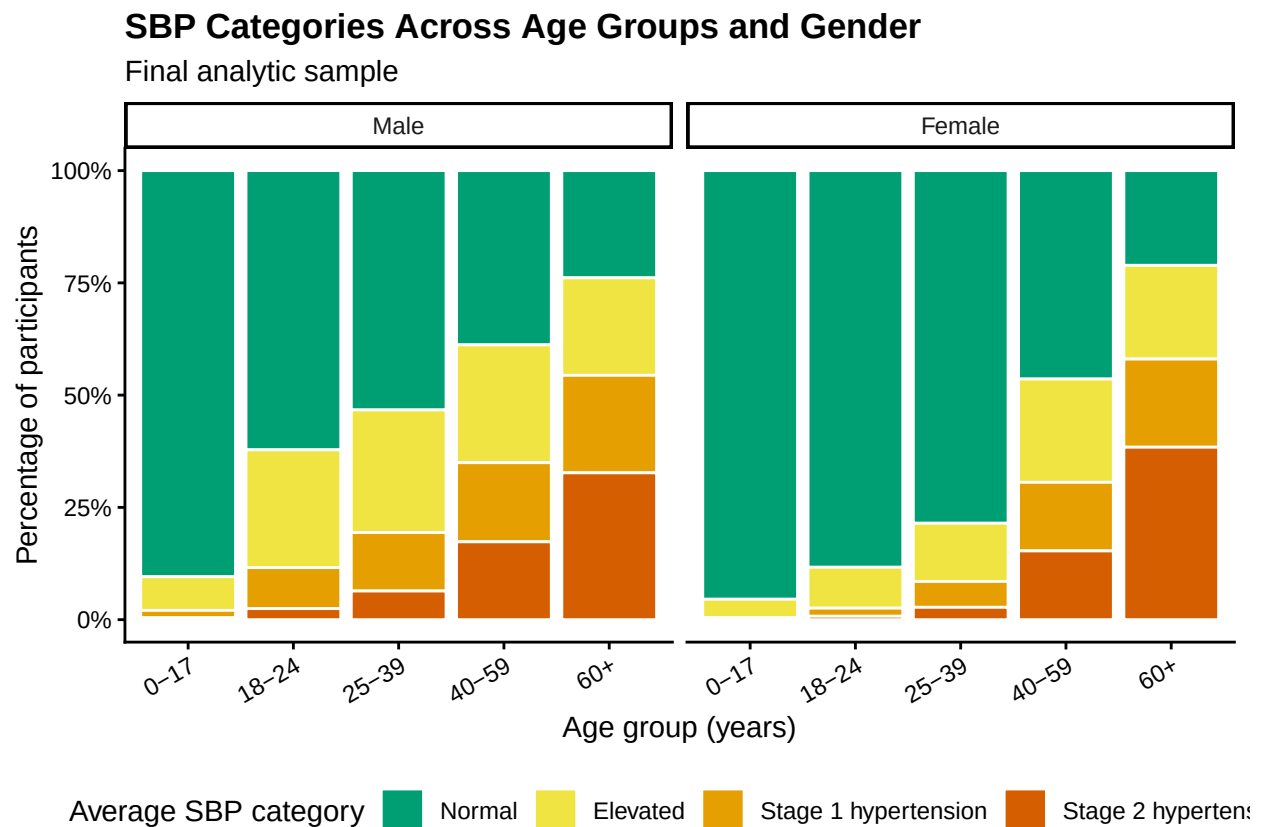
### Required final faceted visualization
plot_sbp_age_gender <- ggplot(
  final_sample,
  aes(x = age_cat, fill = sbp_category)
) +
geom_bar(position = "fill", colour = "white") +
facet_wrap(~ gender) +
scale_y_continuous(labels = scales::percent_format()) +
scale_fill_manual(
  values = c(
    "Normal" = "#009E73",
    "Elevated" = "#F0E442",
    "Stage 1 hypertension" = "#E69F00",
    "Stage 2 hypertension" = "#D55E00"
  )
) +

```

```

labs(
  title = "SBP Categories Across Age Groups and Gender",
  subtitle = "Final analytic sample",
  x = "Age group (years)",
  y = "Percentage of participants",
  fill = "Average SBP category"
) +
theme_classic() +
theme(
  plot.title = element_text(face = "bold"),
  axis.text.x = element_text(angle = 30, hjust = 1),
  legend.position = "bottom"
)
plot_sbp_age_gender

```



##The original NHANES dataset included 29400 participants. After excluding participants with missing age, gender, or average SBP, the final analytic sample included 21604 participants; 7796 participants were excluded.

##Comparison of the original and final age-distribution plots indicates whether the removal of missing observations materially changed the age composition of the sample.

##The final faceted figure shows how average SBP categories vary across age groups and between genders.